

Nakul Chamariya

+91 7021590832 — nakulchamariya74@gmail.com — LinkedIn — GitHub — Upwork

TECHNICAL SKILLS AND TOOLS

Languages: Python, TensorFlow, PyTorch

Data Science: Preprocessing, Feature Engineering, Visualization, Statistical Analysis

Machine Learning: Supervised, Unsupervised, Scikit-learn, XGBoost, LightGBM

Deep Learning: TensorFlow, PyTorch, Keras, OpenCV, Transformers

Gen-AI & NLP: GPT-4, LLAMA 3, Langchain, Crew-ai, Hugging Face, Stable Diffusion, OpenAI Whisper

Computer Vision: Object Detection, Segmentation, OCR, OpenCV, Pillow, Yolo

MLOps & DevOps: Docker, DVC, CI/CD (GitHub Actions, CircleCI), AWS (EC2, S3, Lambda, ECR, ECS)

Web: Flask, FastAPI, Streamlit, Selenium, Beautiful Soup

Databases: MySQL, MongoDB, SQLite, Pinecone

Other: Git, Linux

WORK EXPERIENCE

MySellerCentral

Feb'24 – Till date

Senior Data Scientist

Mumbai, India

Skills Used: AWS, Deep Learning, Computer Vision, NLP, Gen-AI, Web Scraping, MLOps, API

- Led end-to-end **infrastructure setup** for production and development environments, implementing structured GitHub repositories with dedicated stage, dev, and prod branches, and developing robust **CI/CD pipelines** using Jenkins and GitHub Actions.
- Designed and managed comprehensive **database architecture** using MongoDB for text data and S3 for media storage, implementing automated daily and weekly backup systems via AWS cronjobs.
- Implemented a scalable **containerization strategy** with Docker utilizing multi-stage builds and Alpine images, achieving **5x cost savings** and **10x latency improvements** across all services.
- Orchestrated complete **cloud migration** of AI modules to AWS Lambda and serverless GPU instances on RunPod, resulting in **10x cost reduction** and monthly savings of **thousands of dollars**.
- **Quantized and deployed Indic-TTS (Indian text-to-speech)** Hugging Face models to serverless GPU infrastructure and configured **AWS VPC** to enhance system security.
- Established comprehensive **monitoring and alert systems** using **Grafana and Prometheus**, including API key exhaustion alerts for various third-party services via AWS Secret Manager and implemented **LLM monitoring** tools to track model responses.
- Built high-performance **ETL pipelines** using PySpark and AWS Glue, reducing processing time for 10TB datasets from **24 hours to 60 minutes**.
- Developed and deployed **AI-powered content generation modules** utilizing **OpenAI Whisper** and **GPT-4o-mini** to create optimized product listings across Amazon, Flipkart, and Shopify marketplaces.
- Created advanced **content evaluation system** with grading modules that assess marketplace listings and provide actionable improvement suggestions, serving over **2 million sellers**.
- Engineered a **multimodal multi-agent system** using **CrewAI, GPT-4o Vision, and Claude 3.5** for automated listing verification (image and text) across diverse marketplaces.
- Implemented **OpenAI batch API optimization**, reducing processing costs by **50%** and saving approximately \$1,000 monthly.
- Developed an **attribute-based sentiment analysis** system for product reviews, deploying **Mistral 8x7B** with AWS Bedrock to achieve near real-time processing (**20 minutes to 2 minutes**) through parallel LLM calls with multithreading.
- Designed **AI-based rich media generation** modules for infographics, lifestyle images, and videos, deployed on cost-effective serverless GPU instances.
- Created a **SQL-driven multi-agent sales analytics module** using **LangChain, CrewAI, and OpenAI Agents SDK**, enabling multilingual audio query processing with insightful visualizations and comprehensive reports.

- Built a **Graph RAG based category mapping system** to identify optimal product categories across multiple marketplaces, improving listing accuracy and discoverability.
- Developed a **custom LLaMA 3 8B-based customer support chatbot**, fine-tuned on internal data for seller query resolution, achieving **5x reduction** in inference cost and **10x improvement** in latency through quantization.
- Strategically leveraged both **open-source LLMs** (Mistral 8x7B, LLaMA) and **proprietary models** (GPT-4, GPT-4o-mini, Claude 3.5 Sonnet) across diverse generative AI applications, optimizing for performance and cost-efficiency.

goGlocal Private Limited

Data Scientist

Feb'23 – Feb'24

Mumbai, India

Skills Used: AWS, Deep Learning, Computer Vision, NLP, Gen-AI, Web Scraping, MLOps, API

- Built **web scraping pipelines** and a **real-time price tracker** with **automated email alerts** for actionable insights.
- Designed tools integrating **OCR**, **object detection**, and **Gen-AI** for optimizing product listings and images.
- Developed **sentiment analysis**, **sales forecasting**, and **bidding optimization** models for enhanced marketplace performance.
- Engineered a **multilingual translation tool** for SKUs and **image translation** with OCR for compliance and localization.
- Enhanced chatbots using **Llama2**, **Langchain**, and **conversation history** modules for compliance and e-commerce support.
- Created **APIs** with **Celery** and **Redis**, along with setting up a **CI/CD pipeline** for scalable deployment.
- Designed a **strategic insights dashboard** to provide category-level marketplace analytics.

Edvak Technologies INC

Data Scientist

Dec'21 – Jan'23

Hyderabad, India

Skills Used: AWS, Machine Learning, Deep Learning, NLP, Speech-to-Text, MLOps, API

- Revolutionized **lab test and medication recommendations** with a **multi-label classification** system.
- Enhanced text accuracy with **Diagnosis Prediction** using **Clinical-Bert** and **Clinical-NER** for medical coding.
- Improved **Speech-to-Text** precision with a custom **spell checker**.
- Developed a **patient attendance prediction** model for clinic show-up insights.
- Streamlined information with a **coreference resolution** model.
- Predicted outpatient visit volumes using a **time-series forecasting** model with **Facebook Prophet**.
- Anticipated consultation durations with a **service time prediction** model.
- Orchestrated **deployment** to **AWS ECR** through Dockerization for scalable access.

ACHIEVEMENTS

- **100 percent success rate** on **upwork** with **5 star ratings** from all clients